

customer-friendly.

The company offers variable capacity machines to cater to several applications such as irrigation, army tanks, construction sites, and more. The startup has introduced multiple variants of AWGs, including VJ-30 LPD unit (small-scale generator), VJ-100 LPD unit (domestic purposes), VJ-400 LPD unit (commercial purposes),

and VJ-2000 LPD unit (solar-powered off-grid settlements). For these units to work properly and efficiently, the temperature should lie between 20 and 42 degrees Celsius with a relative humidity of 35-99 per cent. The weight and size of these units vary based on the requirement for the rated production capacity.

Devices that have been installed

in various places like schools, restaurants, houses, offices, and so on, have already dispensed more than 85,000 litres of water. The team is also working on a VJ-1000 LPD unit, and with its expansion to more cities and other deprived regions, the demand-supply gap for potable water will gradually diminish.

—Ayushee Sharma

5 THE IoT FOR DIGITAL FARMING

Agriculture is one of the most important industries in the world that not only provides food but also raw materials for other industries like textiles. Still, it is plagued by several issues, both natural and man-made, that make life difficult for the farmers.

xFarm, an agtech startup in Milano, was founded in 2017 out of Matteo Vanotti's need to use a digital service to manage his family's agricultural company in an efficient manner. He was joined by Salvatore Ferullo, the company's CEO and Martino Poretti, head of Research and Development. The company aims to create a comprehensive digital platform using the Internet of Things (IoT) to help farmers and others save time while working.

The ecosystem provided by them includes an agricultural application that works both on mobile and desktop platforms, dashboard analytics to manage multiple farms, and a line of connected IoT sensors optimised for agricultural use that can communicate via the Internet directly with users' accounts.

xSense weather stations and sensors that are necessary to track weather data and environmental parameters come with SIMs so that users can send data via GSM to the software. The application provides up to thirteen free functions like recording treatments and processes in the field, tracking movements and transport, viewing the position of fields and crops, managing warehouse inventory, among others.

Premium xNode smart modules, when paired with sensors, add

advanced functionalities, including forecast models, telemetry, finance management, and precision fertilisation for improved work efficiency and productivity. As the modules use a LoRa wireless system, it is easy to connect up to a hundred devices within an approximate range of ten kilometres on the same network without high battery consumption.

This enables farmers to track activities and analyse local and reliable data, monitor crops, obtain personalised recommendations for their plots to plan optimal crop schedules, improve irrigation and fertilisation, protect plantations, and more. The information provided under the environment tab is made simple to understand with the help of graphs and images. Using telemetry, it becomes simple to control the performance of vehicles, send prescription maps, and allocate fuel expenses.

Besides farmers, target customers include associations, cooperatives, contractors, dealers, agronomists, insurances, and the like. For those employing several workers for their farms, staff management options are available to streamline the process. These customers can keep track of market prices for different crops as well as analyse and manage their company's financial statements.

The mobile version available for Android version 5.0 and above is free without in-app purchases. Sensors and advanced modules have to be bought. In the past few years, many other IoT-based startups like Auroras, MotorLeaf, Swinesmart, FieldIn, and Cowlar have also started investing in agriculture and dairy industries. The ultimate goal is the same, that is, to aid people in this sector and thereby boost the economy and well-being of individuals. **EFY**

—Deepshikha Shukla

